9 **(a)** Define the mass defect of a nucleus.

....

....

.... [2]

(b) Thorium-228 ($^{228}_{90}Th$) is radioactive.

Table 9.1 shows the mass defects of thorium-228 and some other nuclides.

Table 9.1

nuclide	mass defect / u
$^{4}_{2}He$	0.030 377
$^{224}_{88}Ra$	1.846 840
$^{228}_{89}Ac$	1.869 820
$^{228}_{90}Th$	1.871 290
$^{228}_{91}Pa$	1.866 129

(i) Use the data in Table 9.1 to explain why thorium-228 is an α-emitter and **not** a β^- emitter or a β^+ emitter.

....

....

.... [4]

(ii) Calculate the energy, in MeV, released by the α-decay of one thorium-228 nucleus.
Give your answer to three significant figures.

energy = ... MeV [3]

[Total: 9]